



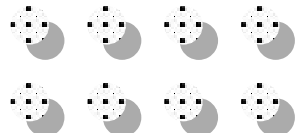
Bathroom Pro 2000

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02/23/26

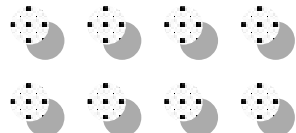
Planning

- Main integrations should work
 - model
 - preprocessing
 - zigbee integration
- POC prioritized
 - integration/optimizing later

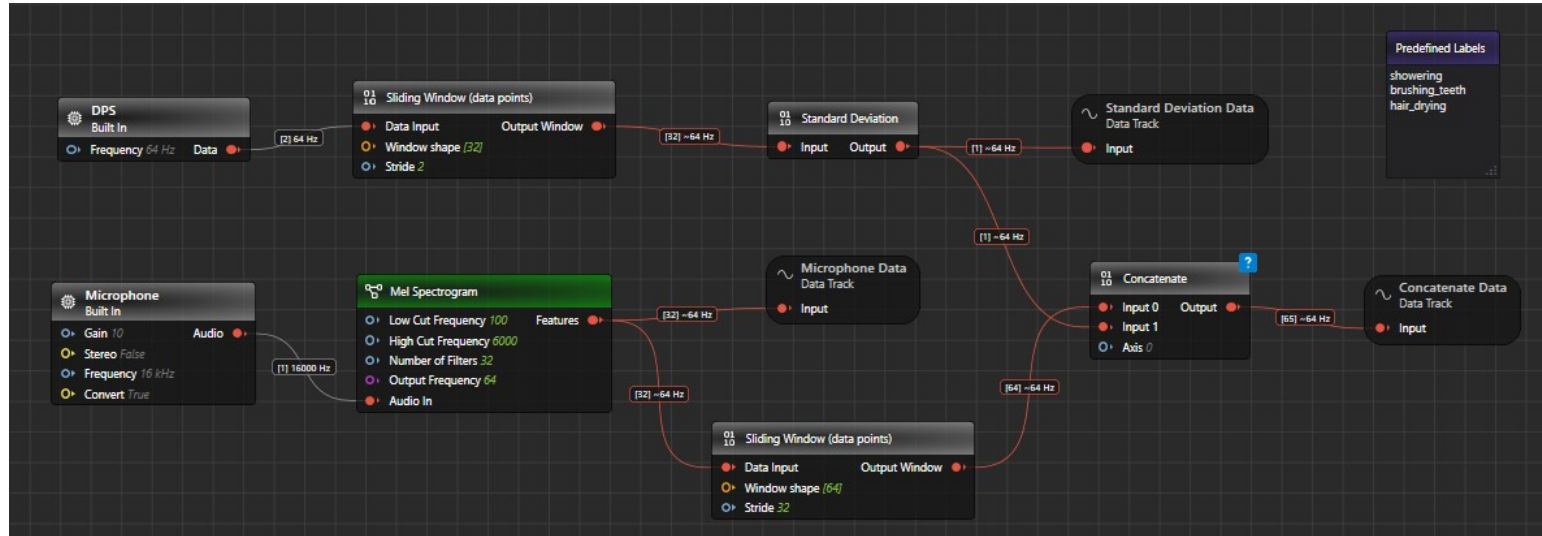


Our workflow

- as we are roommates
 - lots of pair programming
 - 4 eyes see more than 2
 - discussions at dinner



Preprocessing

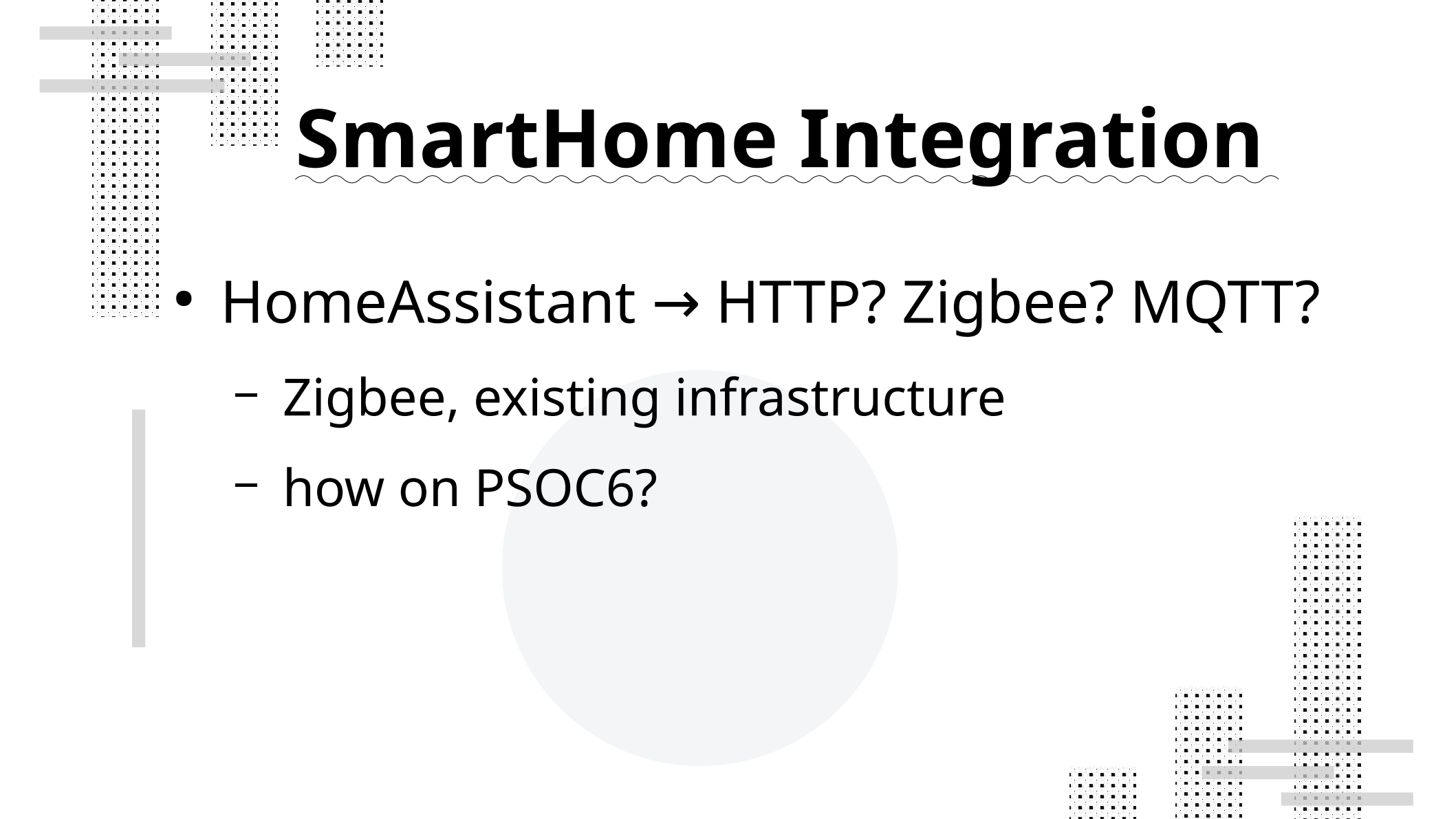


Build a preprocessor by combining different preprocessing layers.
The preprocessor manipulates all data before it is input into models.

Input Shape 

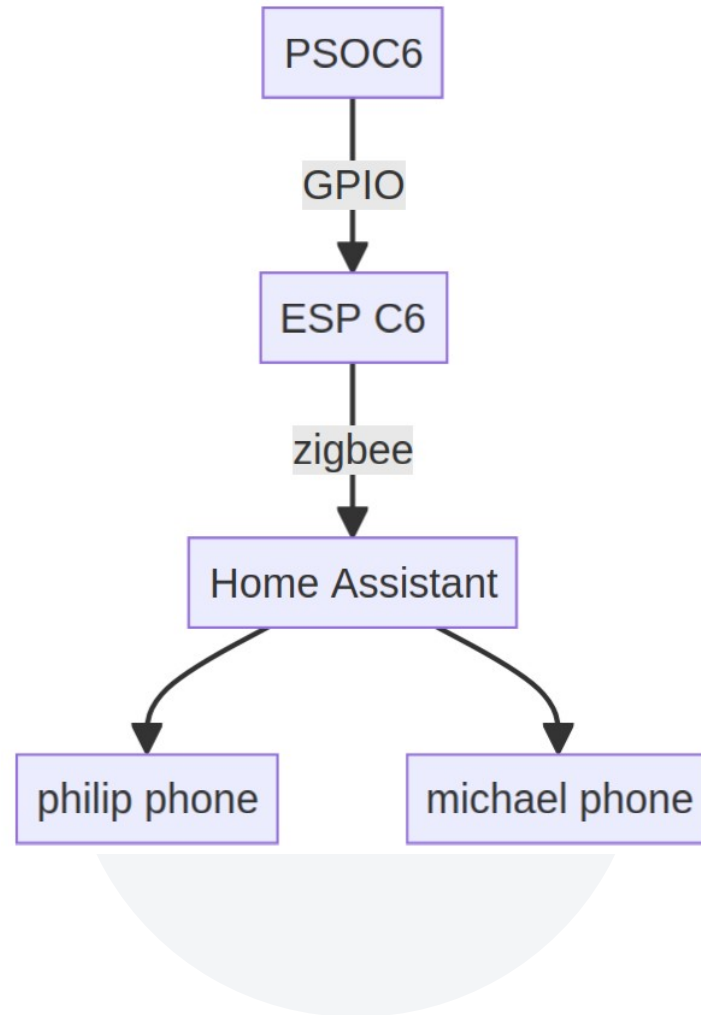
Input Frequency Hz 

Name	Shape	Frequency	Rate		
Sliding Window (data points)					
Window Shape: [512] Stride: 160 Buffer Multiplier: 1	[512]	@ 100 Hz	204.8 KB/sec		Properties
Hann smoothing					
Symmetric: True	[512]	@ 100 Hz	204.8 KB/sec		Properties
Real Discrete Fourier Transform					
Axis: 0	[257,2]	@ 100 Hz	205.6 KB/sec		Properties
Frobenius norm					
Axis: 0	[257]	@ 100 Hz	102.8 KB/sec		Properties
Mel Filterbank					
Number of Filters: 20 Sample Rate (Hz): 16000 Low Frequency Cutoff (Hz): 300 High Frequency Cutoff (Hz): 8000 HTK formula: True Librosa formula: False	[20]	@ 100 Hz	8 KB/sec		Properties
Clip					
Min: 0.000316227766016 Max: 3.40282347E+38	[20]	@ 100 Hz	8 KB/sec		Properties
Logarithm					
Logarithm base: 0	[20]	@ 100 Hz	8 KB/sec		Properties
Sliding Window (data points)					
Window Shape: [60,20] Stride: 660 Buffer Multiplier: 1	[60,20]	@ 3.03 Hz	14.55 KB/sec		Properties



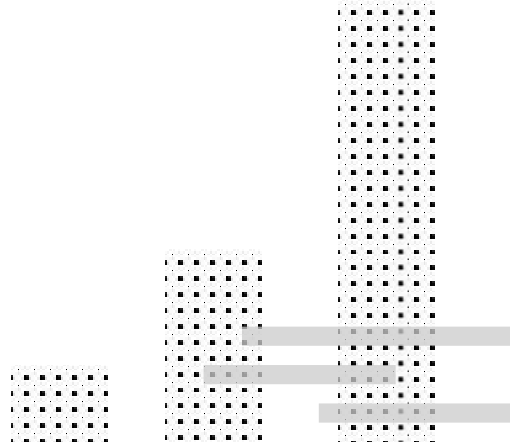
SmartHome Integration

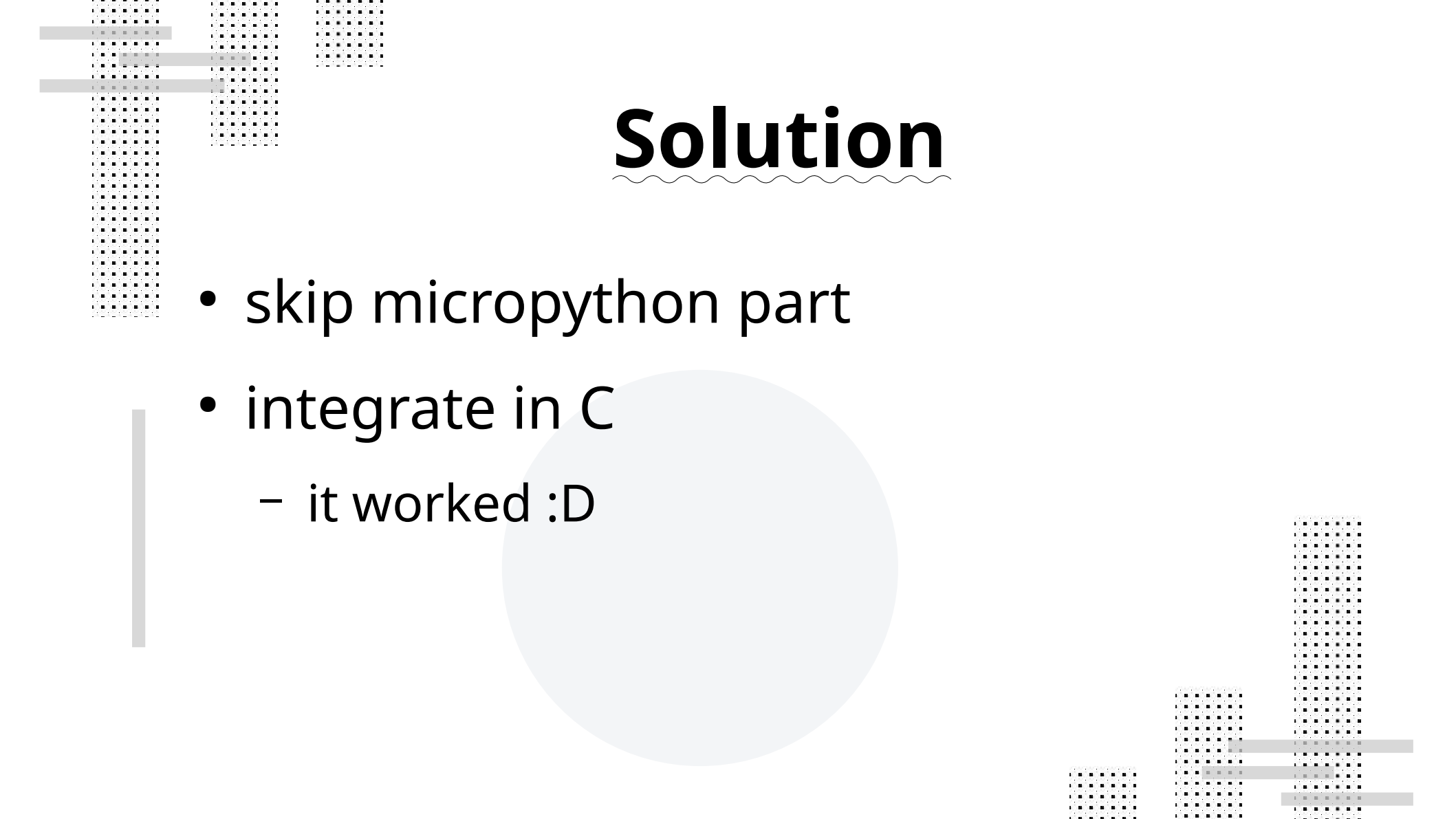
- HomeAssistant → HTTP? Zigbee? MQTT?
 - Zigbee, existing infrastructure
 - how on PSOC6?





Model

- tested lots of models (nearly all variants)
 - in deepcraft studio they work
 - on board they didn't work
- 
- 

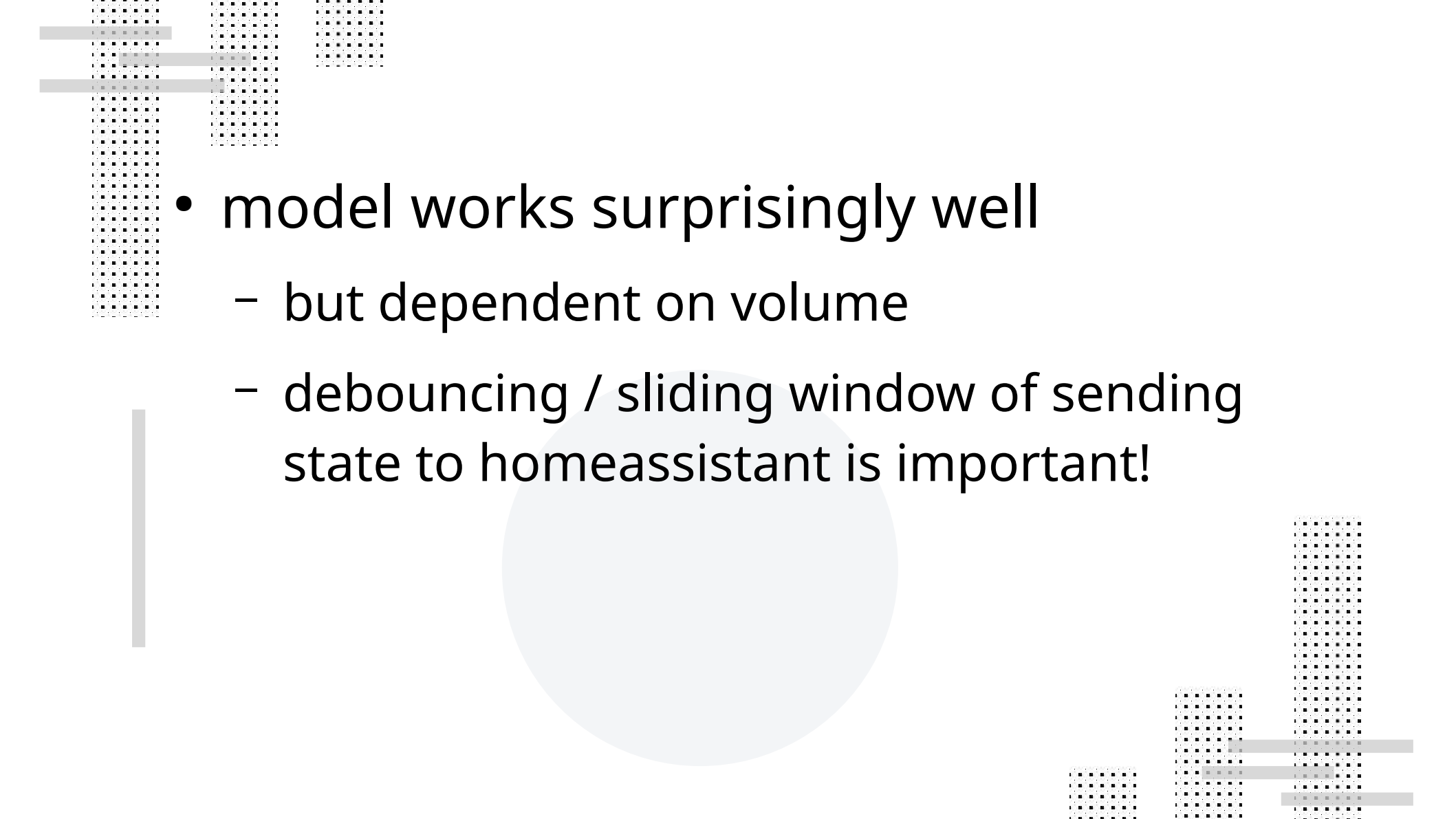


Solution

- skip micropython part
- integrate in C
 - it worked :D

Resume

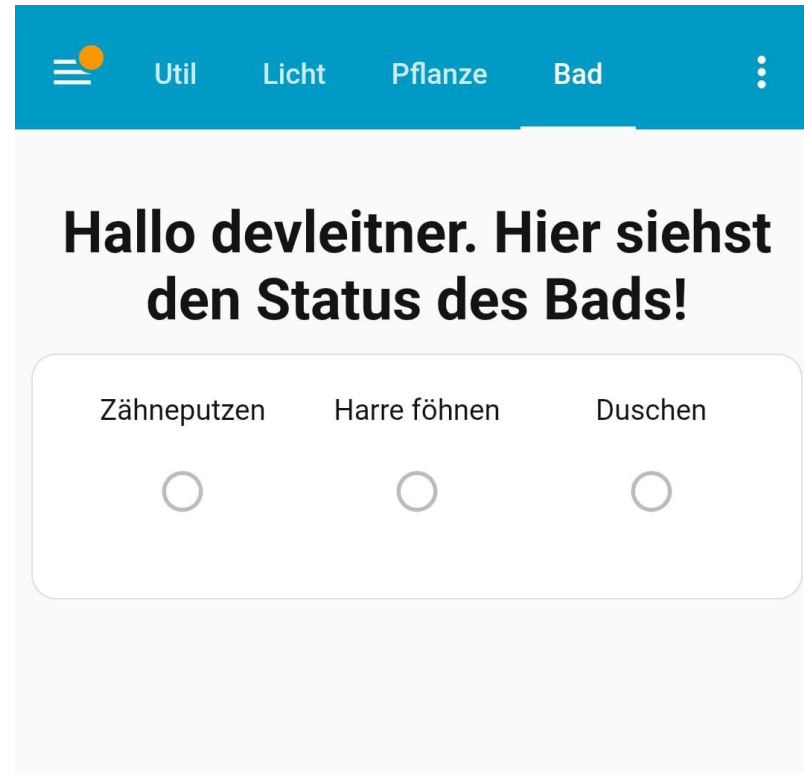
- lots of small undetectable, difficult to trace problems
 - commit in micropython fork destroyed model converting (#6bd82f9)
 - cannot enqueue (why?)
 - never dequeuing (why?)
 - freezes after 3 sec (some buffer overflow?)

- 
- model works surprisingly well
 - but dependent on volume
 - debouncing / sliding window of sending state to homeassistant is important!

The background features several decorative elements: a large light blue circle in the center, a vertical gray bar on the left, and various gray rectangles and dotted patterns in the corners. A wavy line is positioned directly below the text.

Nonetheless still lots of fun!

Result



Hallo devleitner. Hier siehst den Status des Bads!

Zähneputzen

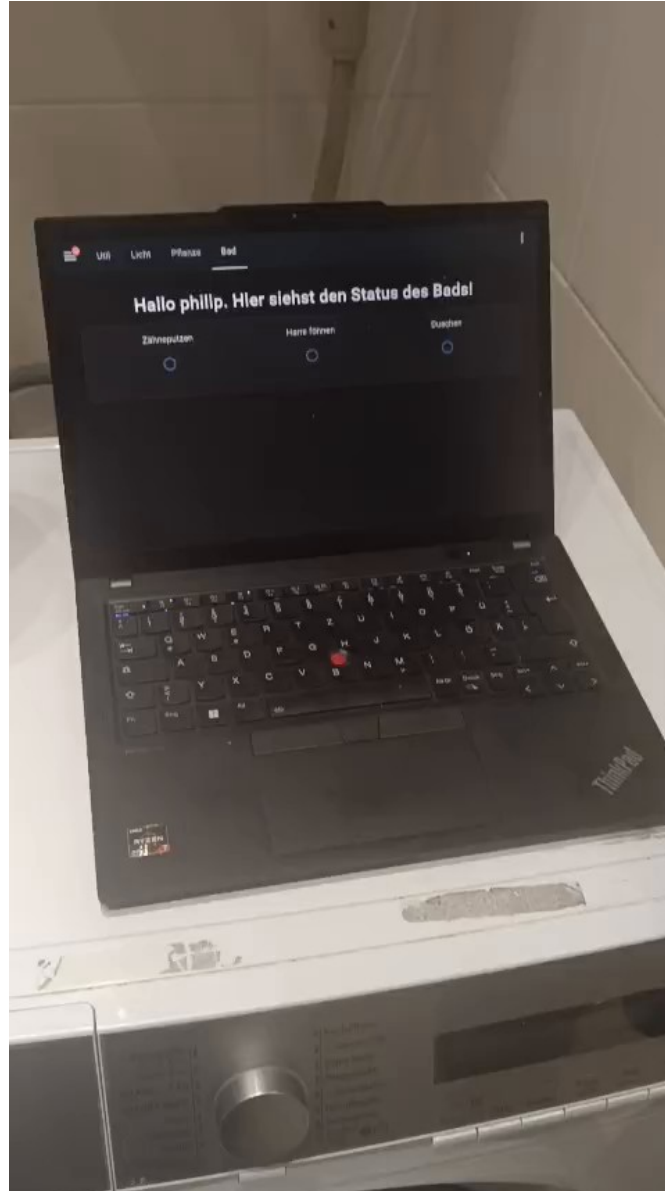


Harre föhnen



Duschen





Thank you!

We are really grateful for the opportunity to work and try out these technologies and work together with you!

